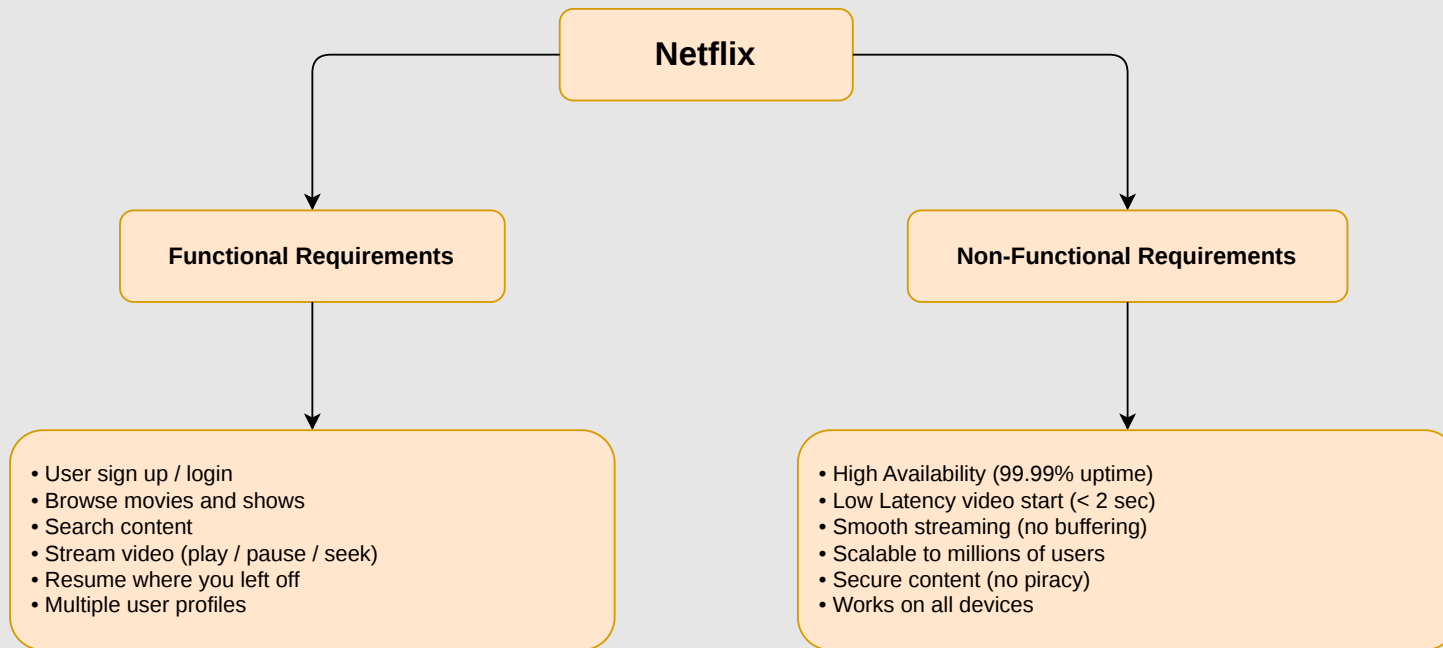
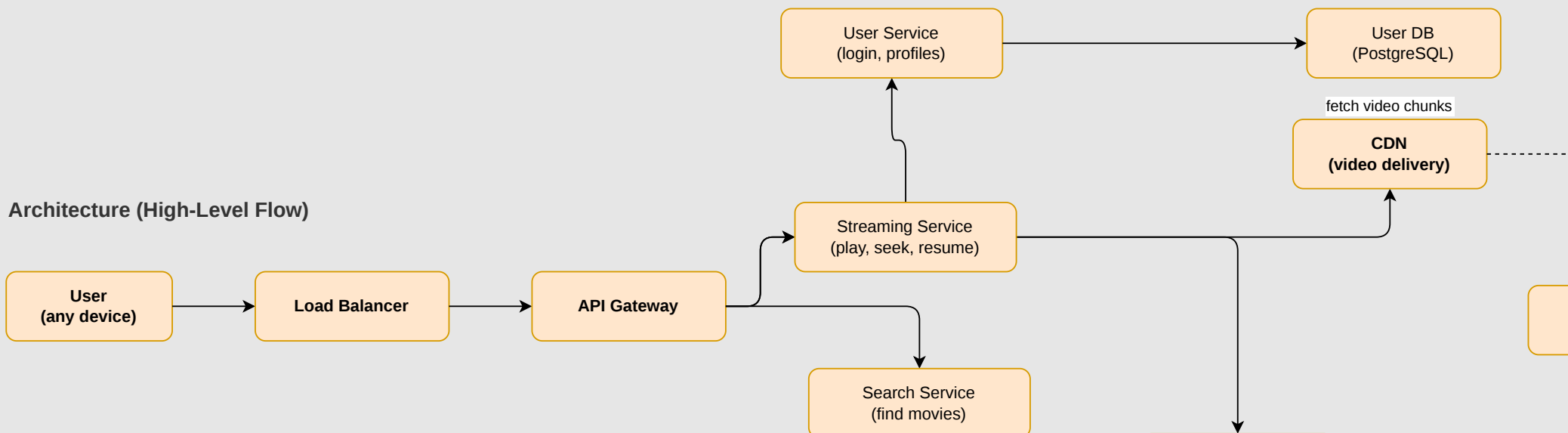




Architecture (High-Level Flow)



Assumptions

- 200 million users worldwide
- 100 million streams per day
- Average video size = 5 GB (per hour, HD)
- Read : Write ratio = 1000 : 1 (mostly watching, rarely uploading)
- Videos stored in multiple qualities: 360p, 720p, 1080p, 4K

Capacity Estimation (Quick Math)

Storage

100M streams x 5GB
= 500 PB total video

Bandwidth

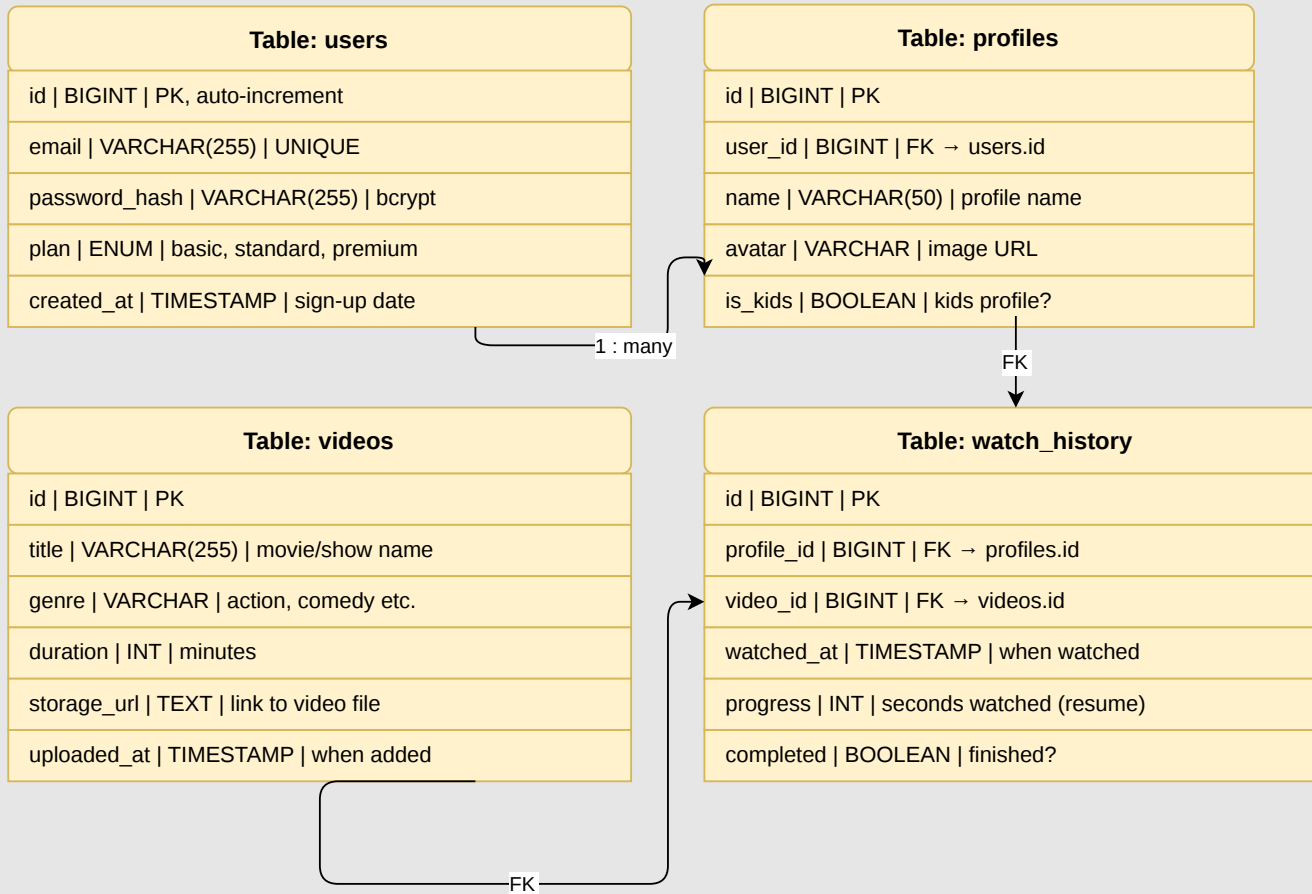
100M streams x 5 Mbps
= 500 Tbps peak

CDN saves bandwidth by serving video from servers closest to the user — no need to go to origin every time.

Video Storage
(S3 / Blob store)

cache miss → origin

DBMS Schema



Cache
(Redis)

check cache first

How Video Streaming Works (step by step)

